Paper id: 250631

Roll No: | | | | | | | | | | | | | |

**BTECH**
**(SEM VI) THEORY EXAMINATION 2024-25**
**COMPILAR DESIGN**

**TIME: 3 HRS** **M.MARKS: 70**

**Note:** Attempt all Sections. In case of any missing data; choose suitably.

## SECTION A

**1. Attempt *all* questions in brief. 02 x 7 = 14**

| Q no. | Question |
|---|---|
| a | Differentiate between phases and passes of a compiler. |
| b | State any two differences between top-down and bottom-up parsing. |
| c | Mention the difference between inherited and synthesized attributes. |
| d | Explain the role of a hash table in implementing a symbol table. |
| e | What is the Finite State Machines (FSM)? Design a FSM that accepts all strings of *a* and *b* containing even number of a's and odd number of b's. |
| f | Write a note on Lexical and Syntactic structure of a programming language. |
| g | What is machine-independent code optimization? |

## SECTION B

**2. Attempt any *three* of the following: 07 x 3 = 21**

| | |
|---|---|
| a. | Explain in detail the process of compilation. Illustrate the output of each phase of compilation of the input ***"a = (b + c)*(b + c)* 2".*** |
| b. | Using rightmost derivations or the derivation trees show that the grammar with following production rules is ambiguous<br>S ⟶ aSbS \| bSaS \| ε |
| c. | Analyze the role of symbol tables in different phases of compilation. How does symbol table interact with lexical, syntax, and semantic analyzers? |
| d. | Given the grammar:<br>S → Aa \| bAc \| Bc \| bBa<br>A → d<br>B → d<br>Construct the **SLR parsing table** and determine whether any conflicts occur. |
| e. | Semantic errors are often difficult to detect. Explain different types of semantic errors with examples and how a compiler detects them during semantic analysis. |

## SECTION C

**3. Attempt any *one* part of the following: 07 x 1 = 07**

| | |
|---|---|
| a. | Construct LL(1) parsing table for the following grammar<br>S ⟶ aB \| aC \| Sd \| Se<br>B ⟶ bBc \| f<br>C ⟶ g |
| b. | Explain the major design issues in code generation. How do these issues impact the performance of a compiler? |

**4. Attempt any *one* part of the following: 07 x 1 = 07**

| | |
|---|---|
| a. | Parse the input string (a,(a, a)) using shift reduce parser<br>S ⟶ (L) \| a<br>L ⟶ L, S \| S |
| b. | Discuss the optimization techniques used at the basic block level. Explain with an example how a basic block can be optimized. |

**5. Attempt any *one* part of the following: 07 x 1 = 07**

| | |
|---|---|
| a. | Write Quadruple, Triple, and three address code for the expression (a/b)*(a+b) - (a*b)/d |
| b. | Give various types of three address statements used for different programming language constructs such as assignment, jump, procedure calls, array statements |



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| | |
|---|---|
| | etc. |

**6. Attempt any *one* part of the following: 07 x 1 = 07**

| | |
|---|---|
| a. | What are Static and Dynamic Scope Rules? How are the activation records used in implementation of recursive calls of a procedure/function. |
| b. | Generate a three address code for the following code segment<br>c = 0;<br>while (c<5)<br>{<br>if (a < b) then<br>x++;<br>else<br>goto p;<br>c++;<br>}<br>p: printf(”Not Done”); |

**7. Attempt any *one* part of the following: 07 x 1 = 07**

| | |
|---|---|
| a. | What are the principle sources of optimization of code? How the DAG is used in common sub-expression elimination? Discuss in details. |
| b. | Give a classification of various errors and discuss each class of errors with examples. |

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